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ORIGINAL EMPIRICAL RESEARCH

A structural model of purchases, returns, and return-based targeting strategies

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Abstract
Despite the importance of product returns, several less well understood issues still remain. In this paper, we propose a dynamic structural model that characterizes consumers' sequential shopping trips for purchases and/or returns. We calibrate our proposed model to individual panel data in which we observe each consumer's sequential shopping trips, including the number of products purchased and/or returned at each trip. Through our simulations, we find that product returns can also lead to desirable outcomes for retailers and that the ability to return purchased products incentivizes consumers to make more shopping trips and thus purchase more. Motivated by these observations, we also examine two possible return-based targeting policies: (1) targeted reduction of return cost and (2) targeted price promotion on return trips.

Keywords Product return · Targeting · Dynamic structural model · Perceived value · Store traffic

Introduction

Product returns have become increasingly problematic for retailers worldwide. In 2018, product returns accounted for $369 billion (or nearly 10%) of total revenue among U.S. retailers (Appriss Retail 2018). This metric may be even higher for certain firms. For example, product returns cost L.L.Bean $50 million annually, accounting for roughly 30% of the firm's profit (O'Connor 2015). Petersen and Kumar (2015) also demonstrate the popularity of product returns among consumers, with more than 60% of consumers returning a purchased item in the apparel, high-technology, and general merchandise product categories. Despite the vast loss in sales from product returns, retailers seem reluctant to entirely eliminate them. In practice, the majority of retailers respond just by slightly tightening the return policies. L.L.Bean, for example, discontinued its extremely liberal return policy (with no time limit or receipt requirement) in February 2018. Now, the retailer requires that all products are returned within a year of the purchase date (Turner 2018). However, even this revised policy is lenient, offering consumers considerable flexibility in returning products.

Why do retailers still have product return policies if they are so costly to maintain? A plausible explanation is that product returns offer additional value to retailers to engage and incentivize consumers for additional purchases. For example, Bechwati and Siegal (2005) claim that strict return policies can decrease consumer purchases. Other studies also show that restricting consumer returns might not be optimal (e.g., Abbey et al. 2018).

In this research, we aim to understand the possible benefits associated with product returns. In particular, we focus on two effects of product returns. First, our proposed model quantifies the "option value" effect of product return. Consider, for example, the case of personal care products. While consumers can obtain a considerable amount of product-related information in-store (e.g., product ingredients), they may not be fully aware of some issues related to the product until they have repeatedly used it. For example, possible allergic reactions will not manifest until they have used the product extensively. If they have the option to return a product if allergies occur after usage, they might purchase in the current period even though the chances of developing allergies are not observable in-store. If so, product returns could increase gross sales of the product. Moreover, if allergies were never to manifest, consumers might keep the product rather than return it. Consequently, allowing product returns might even end up

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promoting net purchases (after accounting for losses in sales associated with product returns).

Second, product returns could also impose a separate "shopping trip" effect on consumers' decisions in the future. For example, product returns might promote store traffic as the need to return any products (as an outcome of the current-period purchase) requires making a separate shopping trip (i.e., visit the store again) in the future. Consider consumer A, as observed in our data, who visited a store to return a product purchased previously. During the return trip, she also purchased additional items, apart from the product return. Similar to the option value effect, shopping trips for the purpose of making returns could also potentially lead to additional purchases. We refer to such an effect as the shopping trip effect of product return.

To fully evaluate a retailer's return policy, it is therefore important to empirically quantify the magnitude of these two effects. In particular, retailers need to understand whether and by how much these two effects could increase consumers' net purchases. Unfortunately, empirical evaluation of these two effects is challenging from archival data alone—the effect of product returns cannot be fully realized simply by comparing the purchases or shopping trips of consumers who have returned items and those who have not. Such an analysis would be subject to a selection bias, as consumers who have returned items could be fundamentally different from those who have not (e.g., loyalty, willingness to pay). Thus, behaviors are not directly comparable between the two groups.

One way to address this selection bias may be through randomized field experiments. For example, a firm could temporarily ban product returns from a randomly selected group of consumers (the treated group) and allow product returns from a control group. Because the firm exogenously determines the ability to return products, comparison between the two groups could unbiasedly reveal the impact of product returns. Unfortunately, such a procedure could be costly to implement for retailers, because return policies usually require collective decisions across multiple divisions, including marketing, logistics, and finance. Moreover, changes in return policies could lead to overreaching effects on the customer relationship. For example, consumers might be confused and irritated with even small changes in return policies for short periods. That is, temporarily banning product returns by the treated consumers might jeopardize their relationship with the retailer, which could be costly to restore after the experiment. Subsequently, we further demonstrate that designing an appropriate experiment to measure the effect of product returns is also nontrivial, even if all the challenges can be overcome.

To address these empirical challenges, we propose a dynamic structural model that accounts for consumers' sequential shopping trips, for the purpose of purchases and/or product returns. A structural model enables us to conduct policy simulations to quantify consumers' responses to hypothetical changes in return policies. In contrast with randomized field experiments (in which changes need to be implemented in reality), policy simulations are hypothetically conducted through the proposed model and therefore do not interfere with the firm's return practice. Thus, policy simulations with our proposed structural model are less costly to implement than field experiments.

Our proposed model has two features that are directly related to the two effects of product returns we attempt to measure. First, we allow consumers to be forward-looking when making decisions in each period. To make current-period purchase decisions, consumers will consider the possibility and utility of returning the purchased products in the future. Accounting for consumers' forward-looking behaviors associated with future product returns enables us to examine the option value effect of product returns (Anderson et al. 2009). Second, by modeling consumers' decisions repeatedly over multiple periods, our proposed model also enables us to quantify a separate and understudied shopping trip effect associated with product returns. In the context of retailing, a consumer's need to return a product could also influence the occurrence and timing of his or her future shopping trips. For example, if the ability to return has a specific time frame (e.g., 30 days after purchases), the consumer might need to make an additional shopping trip sooner than anticipated to make a timely product return. Thus, understanding the impact of product returns on shopping trips decisions is important, because although the consumer might make an additional trip with the original intent to return a product, he or she might end up purchasing extra items. By accounting for these two effects, our proposed model enables us to effectively assess a retailer's current product returns policy.

We calibrate our proposed model to consumer panel data from a multinational offline retailer. In the data, we observe the sequential shopping trips each consumer made, as well as his or her purchase and/or return decision on each trip. We apply our proposed model to consumer purchases and returns in a brick-and-mortar setting for two reasons. First, brick-and-mortar stores still account for the majority of retail sales (more than 85%) despite the increased popularity of e-commerce (Ali 2019). Second, many consumers prefer brick-and-mortar to online channels because of the possibility of easier and cheaper returns (Rekuc n.d.). Given the level of revenue from offline channels and consumers' motivations to purchase from brick-and-mortar stores, it is important for offline retailers to understand consumers' return behaviors.

Our simulation results reveal the presence of both the option value and shopping trip effect associated with product returns. We further show that both effects can promote consumer engagement on the dimensions of net purchases and shopping trips. In other words, allowing product returns is ultimately in the retailers' interests. Motivated by these observations, we further explore possible strategies for retailers to

Table 1 Literature review and positioning of this research

Articles	Modeling components			Effects of product return		Forward-looking behavior
	Purchase	Return	Shopping trip	Option value	Shopping trip	
Anderson et al. (2009)	√	√		√		√
Ertekin (2018)	√	√				
Ertekin et al. (2020)	√	√				
Petersen and Kumar (2009)	√	√				
Petersen and Kumar (2015)	√	√				
Sahoo et al. (2018)	√	√				
This study	√	√	√	√	√	√

take better advantage of the marketing opportunities associated with product returns. We consider and evaluate two targeting strategies: (1) targeted reduction of return cost and (2) targeted price promotion on return trips. The first policy increases the expected return utility and, thus, the option value effect of product returns, while the second policy enables the firm to focus on marketing opportunities associated with the shopping trip effect of product returns. Our simulation results suggest that both policies could effectively promote net purchases (after accounting for the losses in sales due to product returns) by 7.52% and 11.51% respectively.

This paper adds to the past empirical work of product returns on the following dimensions. Methodologically, we propose a dynamic structural model to better quantify the effects of product returns. The proposed option value and shopping trip effects focus not only on purchases but also on shopping trips. Both are important metrics that measures consumer engagement with the retailer; our findings therefore allow retailers to better evaluate their return policies. Substantively, using policy simulations, we identify two possible targeting strategies that enable retailers to better leverage the desirable impacts of product returns. In the next section, we provide detailed discussions about the positioning of this paper.

Literature review

This research relates to an established stream of literature on product returns. Methodologically, we build our proposed model on the theoretical underpinnings of prior research. In particular, we consider three factors that contribute to consumers' return decision: product perceived values (Akçay et al. 2013; Altug and Aydinliyim 2016; Gallino and Moreno 2018; Hsiao and Chen 2012; Moorthy and Srinivasan 1995; Ofek et al. 2011; Su 2009), consumer heterogeneity (Akçay et al. 2013; Hsiao and Chen 2012; McWilliams 2012; Su 2009), and transaction cost (Hsiao and Chen 2012; Su 2009).

Table 1 summarizes the positioning of our study to extant empirical research on product returns. In terms of modeling components, prior studies have primarily examined the impact of product returns on purchase decisions (e.g., Ertekin 2018; Ertekin et al. 2020; Petersen and Kumar 2009, 2015; Sahoo et al. 2018; Shulman et al. 2009). In the retail setting, however, consumers' engagement with retailers can be multidimensional and hence extend beyond purchases.

Apart from purchases, the study also examines the impact of product returns on shopping trips. Quantifying the additional shopping trips driven by the needs to return products could be managerially important. Even though these additional trips are made with the original purpose of product returns, retailers might still benefit especially when price promotions could be offered to incentivize purchases during these trips. To formally examine the value of marketing opportunities derived from these shopping trips, we conduct a policy simulation and find that the retailer could gain an 11.6% net increase in revenue by offering price promotions to consumers who make these shopping trips. The "shopping trip" component in our proposed model therefore enables us to examine possible strategies for retailers to strengthen the financial benefits derived from product returns.

Methodologically, our research also differs from the majority of the literature on product returns.[1] In particular, we explicitly model consumers' forward-looking behaviors, drawing from the broader stream of literature on dynamic structural models (e.g., Erdem et al. 2003; Hendel and Nevo 2006; Ishihara and Ching 2019). As discussed previously, accounting for forward-looking behavior is crucial to understand the option value and shopping trip effects of product returns. Another advantage of a dynamic structural model is the possibility to evaluate consumers' behaviors under varying scenarios through policy simulations. The results of such simulations may not only help quantify the magnitude of the

[1] The exception is Anderson et al. (2009), which will be discussed in the next paragraph.

two product returns' effects but also help retailers design related targeting strategies.

Among other research on product returns, our study comes closest to that of Anderson et al. (2009), who conducted the first empirical examination of product returns using a structural model. While both Anderson et al.'s (2009) work and our study adopt a structural model to assess consumers' return decisions, we would like to note the following differences. First, Anderson et al. (2009) primarily aim to understand how the ability to return currently purchased products affects consumers' current-period purchase decisions (i.e., the option value effect). In a typical retail setting, consumers might repeatedly interact with a retailer over multiple periods, and product returns could influence their decisions both in the current period and in the future. In that regard, we model a consumer's dynamic decisions with a retailer repeatedly over multiple periods. As described earlier, this modeling framework allows us to operationalize the effect of product return on future decisions through the shopping trip effect, in addition to its option value effect. Understandings of this shopping trip effect lead to valuable targeting opportunities, with the possibility to increase retailers' revenue by as much as 11.6%.

Moreover, we also deviate away from modeling consumers' purchase and return decisions under a multicategory framework. Such a model specification requires explicitly modeling the consumer utility with each product category. Considering the number of unique product categories carried by each retailer, estimating a multicategory model could be computationally costly. We therefore focus on consumers' total spend during each shopping trip rather than their purchases in each product category. In the model, product-category heterogeneity is implied by the product value signals. Thus, our proposed model bypasses the computational cost to define and estimate categorical utility and, as such, can be easily applied to different retail settings, regardless of the number of categories.

Structural model of shopping trips with purchases and returns

Model overview

In our model, consumers jointly make shopping trip, purchase, and return decisions. During each period, the consumer decides whether to make a shopping trip (e.g., visit the brick-and-mortar store) for the purpose of purchase and/or product return. We model consumers' decisions repeatedly over multiple periods and allow them to be forward-looking in making decisions in each period.

As a key component in our model, the perceived values of products govern a consumer's purchase and return decisions in each period. In line with Anderson et al. (2009), the perceived value captures how well the characteristics offered by the purchased products fit the consumer's individual tastes. This perceived product value could be formed based on information obtained from multiple sources. During each shopping trip, the consumer could obtain some information of product values by examining the products on display. For example, he or she could become aware of the product ingredients from reading the product packaging or in-store advertising. Such information provides a preliminary understanding of the product values, and a higher perceived value based on the information obtained in-store could contribute to the decision to purchase the product. However, this information does not fully reveal the perceived value of a product from every angle. For example, a consumer might not be aware of a potential allergic reaction with a skin care product until using the product repeatedly. Product usage therefore provides additional information and thus could change the consumer's perceived value of the purchased product, resulting in a potential product return if this value turns out to be low.

Following Anderson et al. (2009), we also allow consumers to be forward-looking when making decisions in each period. The modeling consideration is motivated by the inter-temporal dependence introduced by the possibility of making a product return. As described earlier, current-period purchase decisions could potentially lead to future returns and hence changes in future utilities. A rational consumer therefore will account for the possibility and utility of returning the purchased products in the future, when making purchase decisions in the current period. As demonstrated in Anderson et al. (2009), the consideration of a consumer's forward-looking behaviors enables us to consider the option value effect of product returns, as an important angle to assess the impact of product returns on consumer demand. We will demonstrate how we accommodate this effect after introducing our model.

While the option value effect provides an important way to examine the impact of product returns on purchase decisions, product returns might also influence consumers' behaviors on dimensions beyond purchases. In retailing, the need to make a return could influence the occurrence and timing of consumers' future shopping trips. As discussed earlier, the understandings of consumers' shopping trips could lead to important marketing opportunities for retailers through targeting. Toward this end, our model extends Andersen et al.'s (2009) framework by considering the shopping trip effect to capture the impact of product returns on consumers' both shopping trips and purchases with the retailer. Table 2 summarizes the notations and variables used in our model.

Consumer utility

We first discuss a consumer's flow utility during a single-period shopping trip before specifying his or her dynamic decisions across multiple periods. If a consumer makes a shopping trip at time t, his or her flow utility on the focal

Table 2 Notations and variables

	Notation	Explanation
Indicator		
Subscript	i	Consumer index
	t	Time index
	e	Index of each product (out of E_{it} products purchased during the shopping trip at time t)
Superscript	b	Purchase
	r	Return
Choice-specific indicators	G_{it}	Number of products returned during the shopping trip at time t
	E_{it}	Number of products purchased during the shopping trip at time t
Purchase utility	D_{it}	Stock of perceived value from all purchased products during the shopping trip at time t
	d_{ie}	Perceived value of each purchased product
	δ_i	Decay of utility from additional purchased products
	P_{it}	Total price paid
	v_i	Fixed cost associated with each shopping trip
Return utility	R_i	Marginal cost of returning each product
	D_{it}	Stock of perceived value from all returned products during the shopping trip at time t
	$d_{ig,\,r}$	Perceived value of each returned product
	P_{it}^{G}	Total refund from returns

shopping trip is determined by (1) purchase utility U_{it}^{b}, (2) return utility U_{it}^{r}, and (3) a fixed cost related to purchase and return v_i. Mathematically, we can write consumer i's flow utility provided that a shopping trip occurred at time t as $U_{it} = U_{it}^{b} + U_{it}^{r} + v_i$.

Purchase utility We specify the purchase utility, related to E_{it} number of products purchased during a single trip, as $U_{it}^{b} = D_{it}^{\delta_i}(E_{it}) - \alpha_i . P_{it}$, where D_{it} is the stock of perceived value consumer i obtains from the products purchased from the focal retailer. In line with Anderson et al. (2009), the perceived value captures how well the characteristics offered by the purchased products fit consumer i's specific tastes. The parameter δ_i is the decay parameter related to the utility derived from the perceived value of the products purchased (e.g., Nevo et al. 2016). If $\delta_i < 1$, the marginal purchase utility decreases with respect to the stock of the perceived value. Such specification is appealing because the perceived values from additional purchases may generate diminishing utility. Thus, it allows consumers to purchase products with prices considered on each shopping trip. The term $P_{it} = \sum_{E_{it}} p_{ie}$ is the price paid for all E_{it} purchased products (after application of price promotions), and α_i is the utility weight assigned to price.

We define the stock of perceived value D_{it} as $D_{it}(E_{it}) = \sum_{E_{it}} d_{ie}$, where E_{it} is the total number of purchases on the current shopping trip and d_{ie} is the perceived value for a particular product. In other words, our proposed model assumes that the consumer derives different perceived values from each purchased item and that the total perceived value from the entire shopping trip is aggregated across all purchased items. Similar to Anderson et al. (2009), this perceived value d_{ie} captures the consumer's perceived match with a product. When visiting the retail store, the consumer could form such perceived value from the in-store product experience. For example, the consumer could obtain product-related information by reading the product packaging and/or from in-store advertising. This information could shift the consumer's perceived value toward the product.

In our model, we specify $d_{ie} = \exp(d_i + \varnothing_{ie})$, where d_i represents consumer i's true perceived value of the products offered by the firm. The term $\varnothing_{ie}$ is the random variation, which is specific to item e purchased during the current trip and follows $N(0, \vartheta_i)$. This random variation allows the perceived value experienced in-store to be specific to a shopping trip, a consumer, and a product. Such a modeling assumption is reasonable considering the heterogeneity from three perspectives. First, it is reasonable to expect differences across different shopping trips: (1) the way products are displayed in different stores visited by the consumer on various shopping trips and (2) variations in store environment (e.g., friendliness of sales associates; Ertekin et al. 2020). Depending on these differences, the consumer could have entirely different perceptions even of the same products from different stores. Second, perceived values of the same product could also vary across consumers, when holding all else constant. This heterogeneity could be due to consumers' different shopping behaviors. For example, some consumers might pay more attention to product displays and therefore form a different perceived value of the same products. In our model, ϑ_i indicates heterogeneity among consumers. Last, it is also intuitive that different products (e.g., products have different categories) appeal to

consumers differently. Therefore, parameter ϑ_i also allows us to implicitly capture product heterogeneity. In summary, the random variation $\varnothing_{ie}$ captures the difference in consumers' perceived value of different products and across different shopping trips.

Return utility We specify the utility related to G_{it} number of products returned during a single trip as $U^r_{it} = -D^{\delta_i}_{it,r}(G_{it}) + \alpha_i.P^G_{it} + R_i.G_{it}$. When returning products, the consumer can no longer enjoy their perceived value, as captured by $-D^{\delta_i}_{it,r}(G_{it})$, and P^G_{it} is the refund the consumer receives from the returns. Similar to Anderson et al. (2009), we define a marginal cost R_i as being related to each product returned—for example, effort in locating the receipt or credit card statement as well as the cost the consumer incurs from the need to restore the original packaging.

Similar to purchase decisions, we posit that the consumer's return decision depends on the stock of perceived value $D_{it,r}(G_{it}) = \sum_G d_{ig,r}$ associated with all products being returned during the current trip. Here, $d_{ig,r}$ is the perceived value of a particular product r being returned during the trip. This perceived value stems from the consumer's usage experience (after purchasing the product), for which we assume that $d_{ig,r} = \exp(d_i + \varnothing_{ig,r})$. Similar to the perceived value obtained at the purchase stage, $\varnothing_{ig,r}$ is the random variation related to the randomized perceived value of the returned product obtained from the usage experience. We assume that $\varnothing_{ig,r}$ follows $N(0, \vartheta_{ir})$; therefore, ϑ_{ir} captures the variance of perceived value signals each consumer obtained from the usage experience.

Note that $d_{ig,r}$ is different from the perceived value obtained in-store on the purchase occasion, or d_{ie}. The difference between $d_{ig,r}$ and d_{ie} enables us to rationalize the consumer's return decision. For example, the consumer might be attracted (i.e., receive high realization of d_{ie} in-store) to a skin care product because of its pleasing smell and/or attractive packaging. Postpurchase usage experience could uncover some allergy issues, and these lower draws of $d_{ig,r}$ from usage experience might motivate the product return. Thus, it is reasonable to assume that perceived value signals obtained in-store could be systematically different from those obtained from usage experience (i.e., with different ϑ_i and ϑ_{ir}) even for the same product.[2]

Fixed cost To make any purchase and/or product return, the consumer needs to pay a fixed cost to make a shopping trip, defined as v_i. As Erdem et al. (2003) discuss, the fixed cost can be interpreted, for example, as the cost of going to the store, locating the products in the store, waiting in line to check out, and then bringing products either home or back to the store (for return). Regardless of what this entails, a higher fixed cost will, ceteris paribus, lead households to purchase and/or return less frequently and to purchase and/or return in larger quantities at each purchase/return occasion. This fixed cost might drive consumers to group purchases and/or return multiple products on the same trip rather than making separate trips.

Utility from no trip Alternatively, the consumer could choose not to make a shopping trip in the current period. We assume that a consumer's decision not to visit a store depends on his or her overall relationship with the focal retailer, as captured by RFM (recency, frequency, monetary value/past spending) measures.[3] We define the flow utility from no trip as $f(r_{it}, f_{it}, m_{it}) = \alpha^r_i \log(r_{it}) + \alpha^f_i \log(f_{it}) + \alpha^m_i \log(m_{it})$ to capture the nonlinear relationship of the RFM measures.[4] In our model, we include the following RFM measures: (1) the number of weeks from the most recent purchase (recency), (2) the number of shopping trips with purchases made in the recent quarter (frequency), and (3) the number of net spending in the recent quarter (monetary value). We use log-transformation of these three variables to capture the potential nonlinear relationship.

Dynamic decisions

Our model assumes that consumers are forward-looking in maximizing the utility from purchase and return, net of all applicable costs, across multiple shopping trips during the entire sample period. As discussed previously, a dynamic structural model is desirable in our context because it enables a rational consumer to take future consequences into account when making decisions in the current period.

During each period, consumers' decisions depend not only on the flow utility but also on the expected utility from future shopping trips (e.g., to return products purchased in the current trip). Mathematically, the total utility to make a shopping trip today (when making at least one purchase and/or one return, or $E_{it} + G_{it} > 0$) is

$$U_{it} + \beta.EV\left(x_{i,t+1} | E_{it}, G_{it}\right) + \varepsilon^1_{it}, \quad (1)$$

where U_{it} is the flow utility (as previously defined), ε^1_{it} is the random shock that follows a type-I extreme value distribution,

[2] While it is possible that consumers form the postpurchase perceived quality from a sequential set of signals from repeated usages, we are unable to directly incorporate such a specification into our model as we do not observe consumers' usage behaviors. However, our flexible structure in the distribution of the perceived value signal can accommodate the possible averaging across multiple draws of signals. In particular, signal $d_{ig,r}$ could be the average of a set of repeated random signal draws from usage experience (presumably from a different distribution with much greater variance) and ϑ_{ir} the variance of the distribution related to the average of these random draws. This is another explanation for why we need to differentiate the distribution of perceived value signals from purchase and usage.

[3] Alternatively, we can include RFM measures in the flow utility of purchases and returns when a trip is made.

[4] To capture the effect of customer relationships, we can use the RFM measures to enter the utility associated with either the not purchase/return option or the return/purchase option.

and $EV(x_{i,\,t+1}|E_{it}, G_{it})$ is the expected utility from the shopping trip in the subsequent period. As mentioned previously, the expected utility captures the consumer's expected future consequences of his or her current-period decision (i.e., how the purchase decision made today influences the future decision to return a product and make another shopping trip). Next, $\{x_{i,\,t+1}\}$ is a set of associated state variables, which is both consumer- and time-specific. We define the parametric specification of $EV(x_{i,\,t+1}|E_{it}, G_{it})$ toward the end of this section.

Consumers could also choose not to make a shopping trip and thus make no purchases or returns in the current period ($E_{it} + G_{it} = 0$). We define the utility from no trip as

$$f(r_{it}, f_{it}, m_{it}) + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right) + \varepsilon^0_{it}, \tag{2}$$

where $f(r_{it}, f_{it}, m_{it})$ is the flow utility from no trip and ε^0_{it} is the random shock that follows a type-I extreme value distribution. Similar to the scenario when the consumer makes a shopping trip, the no-trip decision today could be driven by the expectation of making shopping trips in the near future. Including $EV(x_{i,\,t+1}|E_{it}, G_{it})$ in the no-shopping-trip decision allows such interdependence.

Put formally, the consumer makes a decision to maximize the following objective function during each period:

$$V\left(x_{it}|E_{i,t-1}, G_{i,t-1}\right) = \max_{\{E_{it}, G_{it}\}} \left\{ \begin{array}{c} U_{it} + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right) + \varepsilon^1_{it}, \\ f(r_{it}, f_{it}, m_{it}) + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right) + \varepsilon^0_{it} \end{array} \right\}, \tag{3}$$

where the first line of the equation represents the scenario when the consumer makes a shopping trip and the second represents the other scenario. Finally, $\{x_{it}\}$ is a set of associated state variables that depend on consumer decisions in the previous period ($E_{i,\,t-1}$, $G_{i,\,t-1}$).

We now define how we specify $EV(x_{i,\,t+1}|E_{it}, G_{it})$ associated with the consumer's dynamic decision in the current period t. As discussed, $EV(x_{i,\,t+1}|E_{it}, G_{it})$ captures the expected utility from subsequent-period decisions conditional on the consumer's choice today (E_{it}, G_{it}). Similar to Eq. (3), we can write $V(x_{i,\,t+1}|E_{it}, G_{it})$ as

$$V\left(x_{i,t+1}|E_{it}, G_{it}\right) = \max_{\{E_{i,t+1}, G_{i,t+1}\}} \left\{ \begin{array}{c} U^b_{i,t+1} + U^r_{i,t+1} + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right) + \varepsilon^1_{i,t+1}, \\ f\left(r_{i,t+1}, f_{i,t+1}, m_{i,t+1}\right) + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right) + \varepsilon^0_{i,t+1} \end{array} \right\}.$$

Given that $\varepsilon^0_{i,t+1}$ and $\varepsilon^1_{i,t+1}$ follow a type-I extreme value distribution, we follow Rust (1987) and derive the closed-form solution of $EV\left(x_{i,t+1}|E_{it}, G_{it}\right) = \ln\left\{\sum_{\{E,G\}} \exp\left(U^b_{i,t+1} + U^r_{i,t+1} + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right)\right) + \exp\left(f\left(r_{i,t+1}, f_{i,t+1}, m_{i,t+1}\right) + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right)\right)\right\}$.

As mentioned previously, part of $\{x_{i,\,t+1}\}$, or the expected realization of the next-period state variables, depends on the current-period decision (E_{it}, G_{it}). Consequently, the decision to make a shopping trip or not in the current period will influence $\{x_{i,\,t+1}\}$ and, thus, the value of $EV(x_{i,\,t+1}|E_{it}, G_{it})$. Consequently, $EV(x_{i,\,t+1}|E_{it}, G_{it})$ takes different values and imposes different influences on Eq. (1) (decision of making a shopping trip) and Eq. (2) (decision not to make a shopping trip).

Effects of product returns

We now discuss the possible effects of product returns derived from our proposed model. As Anderson et al. (2009) demonstrate, the possibility to return current-period purchases in the future could increase the current-period utility and, thus, purchases, a situation we refer to as the option value effect (from the future return opportunities). Our model captures this effect through the *ex-ante* value function $EV(x_{i,\,t+1}|E_{it}, G_{it})$, or the expected utility with future shopping trips, purchases, and returns. As Eq. (1) shows, a consumer's purchase decision in the current trip depends on the *ex-ante* value function $EV(x_{i,\,t+1}|E_{it}, G_{it})$. With the ability to make product returns, consumers might derive positive utility from returning the product if they encounter a lower product value from usage experience, after purchasing the item. As described earlier, the specification that $EV\left(x_{i,t+1}|E_{it}, G_{it}\right) = \ln\left\{\sum_{\{E,G\}} \exp\left(U^b_{i,t+1} + U^r_{i,t+1} + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right)\right) + \exp\left(f\left(r_{i,t+1}, f_{i,t+1}, m_{i,t+1}\right) + \beta.EV\left(x_{i,t+2}|E_{i,t+1}, G_{i,t+1}\right)\right)\right\}$ allows the value of $EV(x_{i,\,t+1}|E_{it}, G_{it})$ to be positively correlated with the (expected) utility from product returns in the subsequent shopping trip, or $U^r_{i,t+1}$. In other words, an expectation of utility associated with a future return, $U^r_{i,t+1}$, could shift a forward-looking consumer's current-period purchase decision by changing the *ex-ante* value function $EV(x_{i,\,t+1}|E_{it}, G_{it})$.

Anderson et al. (2009) show a similar derivation on how to implement this option value effect depending on consumers' expectations.

As discussed earlier, this paper extends Anderson et al. (2009) by considering a separate, shopping trip effect of product return. In our model, a consumer's utility to make a shopping trip depends on the sum of purchase utility U^b_{it} and return utility U^r_{it}, *minus* the fixed cost v_i. In that regards, the need to return previously purchased products (i.e., those turn out to have lower product values from usage experience) leads to a higher utility of return U^r_{it}, which could change U_{it} as well as the decision to make a shopping trip, when holding all else constant.

Estimation

The likelihood function

The likelihood function of the model comprises the likelihood of consumers' decisions related to purchase (E_{it}) and return (G_{it}) quantity. Given the shopping trip decision specified in Eq. (1), the probability of consumer i making a shopping trip in the current period t is as follows[5]:

$$\Pr\left(I_t^r = \{E_{it}, G_{it}\}\right) = \begin{cases} \dfrac{\exp\left\{(r_{it}, f_{it}, m_{it}) + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right)\right\}}{\exp\left\{(r_{it}, f_{it}, m_{it}) + \beta.EV\left(x_{i,t+1}E_{it}, G_{it}\right)\right\} + \sum_{\{E_{it}, G_{it}\}}\exp\left\{U_{it} + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right)\right\}} \\ \dfrac{\exp\{U_{it} + \beta.EV(x_{it}|E_{it}, G_{it})\}}{\exp\left\{(r_{it}, f_{it}, m_{it}) + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right)\right\} + \sum_{\{E_{it}, G_{it}\}}\exp\left\{U_{it} + \beta.EV\left(x_{i,t+1}|E_{it}, G_{it}\right)\right\}} \end{cases}, \quad (4)$$

where I_t^r denotes the joint decision of purchase and return quantity. The first expression in the equation accounts for the case when consumer i does not purchase or return anything at time t ($E_{it} + G_{it} = 0$), and the second expression specifies the probability that the consumer makes E units of purchases, R units of returns, or both ($E_{it} + G_{it} > 0$). When estimating the model, we discretize the joint choice set of purchase and return quantity depending on their empirical joint distribution as observed in the data. By doing so, we can model consumers' purchase and return quantity choices under the discrete choice framework. Such a specification, especially the assumption of the type-I extreme value error term (Eq. 1), is necessary for us to apply contraction mapping to conveniently obtain the *ex ante* value function $EV(x_{it}|E_{it}, G_{it})$ in Eq. (2). The likelihood function for consumer i can be represented by $L_i = \int\prod_t \left\{\Pr\left(I_t^r\right)^{I_t^r}\right\} dF\left(d_{ig,}\, d_{ig,r}\right)$, where $dF(d_{ig}, d_{ig,r})$ is the distribution of perceived values and needs to be integrated out when computing the likelihood function.

Implementation

State variables In our model, the state variables include the (1) number of products available for return, (2) RFM measures, (3) price, and (4) (expected) perceived value signals from purchase d_{ige} and usage experience, $d_{ig,r}$. All four sets of variables could shift consumer decisions and, thus, the value of $V(x_{it}|E_{i,t-1}, G_{i,t-1})$. The first state variable directly influences the possibility and, thus, the utility of product returns. The RFM measures could influence consumer decisions by shifting the utility from making no shopping trips, while price influences consumer decisions through purchase utility. Last, perceived value signals are also important because they capture expected purchase and product return decisions in the future.

Regarding the law of motion, the first two state variables depend on consumers' purchase and return decisions. Figure 1 depicts the relationship between the purchase and return decisions and the first two state variables. In each period, the number of products available for return depends on the number of products available for return carried over from the previous period *plus* the number of products purchased in the current period *minus* the number of products no longer eligible for return (e.g., past the return window). Consequently, product returns made in a given period will decrease the number of products available for return in the subsequent period, while new purchases made in the current period will increase the number of products available for return. Moreover, the number of products available for return will also decrease if some of the past purchases fall beyond the return window (e.g., 30 days after purchase).

Purchases and returns also influence the evolution of the RFM measures. In particular, product returns decrease the monetary value measure (or the cumulative spend with the firm), while purchases increase the focal measure. As recency captures the duration from the most recent shopping trip with purchases, purchases in the current period will reset (and thus reduce) this measure. Last, additional trips with purchases increase the frequency measure of the focal consumer.

Following prior research (Erdem et al. 2003), we assume that product price in a given period depends on that in the last period when estimating the evolution of price series. While consumers are not aware of the exact level of product value signals, they could form their expectations of these two signals based on the distributions of the two signals.

Estimation routine We estimate the model using Markov chain Monte Carlo. The computational burden of estimating a dynamic optimization model in the presence of continuous state variables and consumer heterogeneity can be significant. Consequently, we adopt the estimation method Imai et al. (2009) propose (IJC algorithm hereinafter). Compared with other estimation routines such as that of Rust (1987), the IJC algorithm offers several computational advantages. The

[5] We discretize the choice set of $\{E_{it}, G_{it}\}$ using the empirical distribution observed in the data by combining infrequently chosen purchases and return quantities in the same bucket.

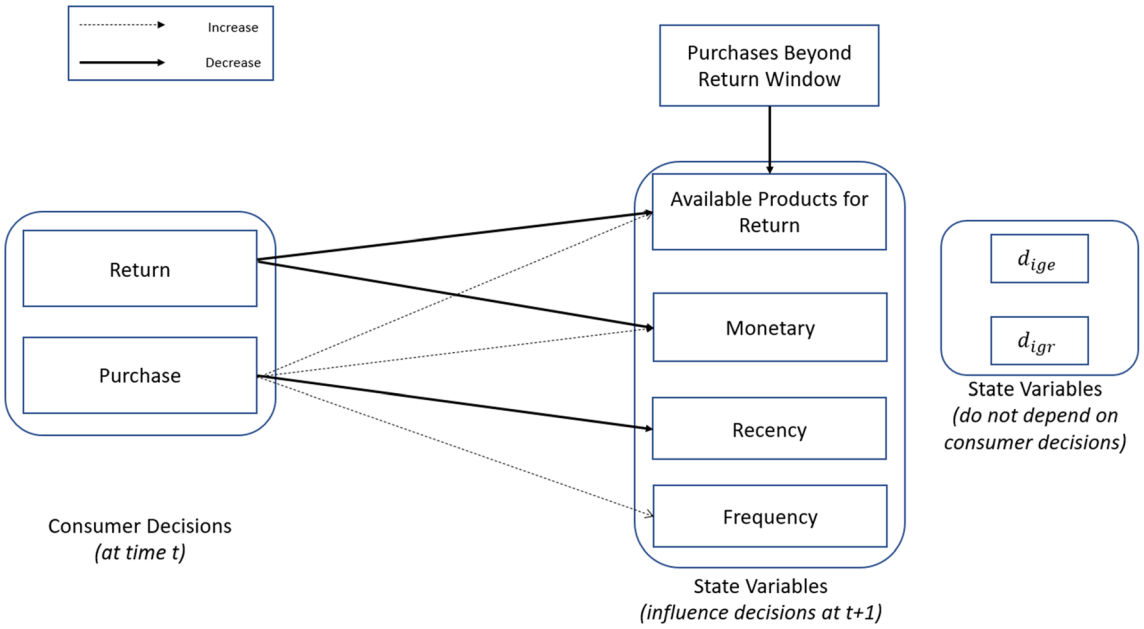


Fig. 1 Consumer decisions and state variables

greatest computational gain comes from eliminating the use of contraction mapping to recover value functions. Instead, the IJC algorithm only requires the value functions to be evaluated once during each iteration. When computing the choice probabilities, *ex-ante* value functions are imputed from past realizations of the value functions using kernel approximations. In addition, the IJC algorithm directly accommodates continuous state space and consumer heterogeneity with only marginally higher computational costs than a homogeneous dynamic model with discrete state variables. In our model, we allow all model parameters to be individual specific; by contrast, a standardized dynamic estimation approach (e.g., Rust 1987) would require us to apply contraction mapping with each consumer's individual parameters at every single iteration, which would be extremely computationally costly. We discuss the detailed estimation procedure in the Appendix.

Data

We calibrate the proposed structural model to consumer panel data from an international brick-and-mortar retailer, whose products are in personal care (e.g., skin care, cosmetics). Our chosen context is well-suited to our proposed theoretical models. In our model, consumers make occasional product returns because of the discrepancy in perceived value signals obtained at the purchase and product usage phase. That is, a consumer might obtain a high perceived value signal in-store but receive a lower-match signal from product usage, which leads to a product return. This discrepancy in perceived value signals is quite likely for personal care products because some product benefits or other issues are challenging to assess in a relatively short period (i.e., in-store). One example is the case of product allergies, as discussed earlier. Furthermore, the vast number of products and the frequent introduction of new products can add to the discrepancy in perceived value signals between the search and purchase stages.

The focal firm allows consumers to return any purchased items (including those purchased on sale) within 30 days from the date of purchase. Consumers can also return the products they have already used. Consequently, they can determine their perceived values with the purchased products from the postpurchase usage experience (as defined in our theoretical model) to decide whether to return the products or not. To return the products, consumers need to visit any of the physical stores operated by the focal firm, and refunds are issued in the original form of payment. The requirement of store visits for product returns is also well-suited to our modeling assumption. During our data period, the focal firm did not impose any charge associated with returns (e.g., restocking fee).[6]

In our data, we observe individual-level purchase and return decisions for a set of 1500 consumers over a two-year sample period. During the period, we observed detailed behaviors of each consumer at the individual level, including the number of items purchased and returned and total spend.

Data description

Figure 2 presents the distribution of shopping trips made by consumers over the sample period. On average, each consumer makes 0.61 trips each month. This relative sparsity observed in our data suggests that consumers incur considerable fixed costs related to purchases and returns. It is therefore

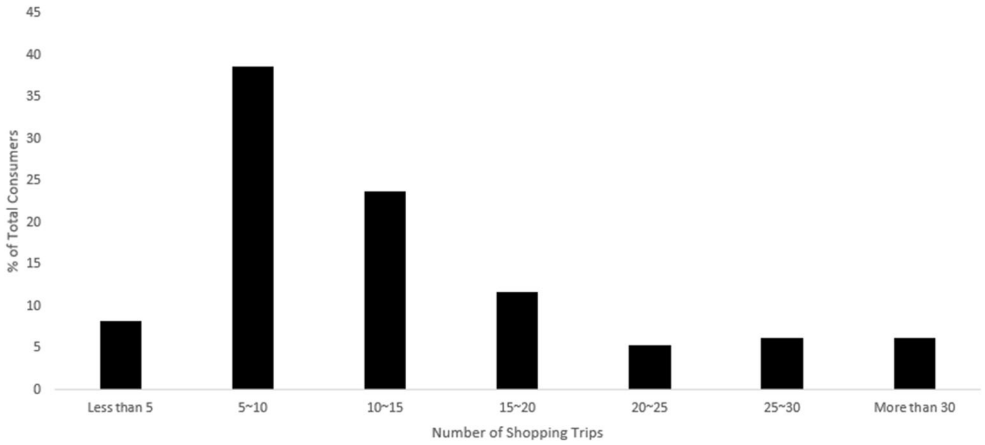


Fig. 2 Distribution of shopping trips

important for retailers to quantify these fixed costs and to understand their influence on consumers' overall interactions with the firm. In addition, given that each consumer makes one trip on average every 1.64 months (=1/0.61) and returns need to be made within 30 days after purchase, the need to return previously purchased items could incentivize consumers to make additional trips sooner; otherwise, previously purchased items could become ineligible for return. These observations motivated us to consider the effect of product returns on additional shopping trips. We empirically examine this conjecture in the next section. In terms of purchase decisions, consumers purchased 74.6 items over the entire sample period, with roughly 11% of the purchases being returned.

Descriptive analysis

We previously discussed the theoretical effects (option value and shopping trip) of product returns derived from our proposed model. We now present empirical evidence that supports our conjectures. We also provide important insights into how these derived effects and associated parameters can be identified in our data.

Number of shopping trips Regarding the shopping trip effect of product returns, we examine the correlation between the number of product returns and the number of shopping trips each consumer made. We find that this correlation is strictly positive and significant (0.21, $p < .0001$). This observation validates our conjecture that consumers make additional trips because of the need to return products.

Return and total spend To examine the overall relationship between return and total spend, in a t-test we compare the *net* total spend (after refunds issued for product returns) between the two samples: consumers who made at least one product return versus those who never returned any products in the same period. We find that the former group tends to spend more (diff=$288.5; $p < .0001$). Such observations could be explained by the option value effect (see Anderson et al. 2009). That is, the possibility of product returns offers consumers additional option values that could increase consumer utility and likelihood to purchase.

In this section, we present reduced-form analyses to demonstrate the effect of product returns. We show that product returns could potentially lead to desirable outcomes. First, product returns can impose a shopping trip effect. That is, returns might incentivize consumers to make additional shopping trips. Even on trips originally meant for product returns, consumers may still make additional purchases. Second, we find that those who make at least one product return also tend to spend more.

Despite the important insights derived from this reduced-form evidence, these analyses alone are not sufficient to causally quantify the impact of returns, due to a potential selection bias. In our setting, such a selection bias might exist at both the consumer and shopping trip level. Consumers who made at least one product return might be fundamentally different from those who did not. Similarly, even for the same consumer, his or her shopping trips with product returns might also differ from trips without. Consequently, consumers or shopping trips with product returns are not directly comparable to those without. A reduced-form approach that does not endogenize consumer decisions could therefore lead to a biased understanding of consumers' behaviors. More important, a reduced-form model is not able to assess alternative policies associated with product returns. Understanding and comparing these policies, however, is important for retailers to explore optimal strategies to handle product returns. Motivated by both the findings and the limitations of the reduced-form approach, we estimate our proposed model next. Simulations based on our proposed model enable us to formally quantify the effect of product returns.

Identification

The parameters to be estimated are (1) shopping trip fixed cost (v_i), (2) parameters related to purchase utility (i.e., mean (d_i)

and variance of perceived value signals in-store (ϑ_i) and from usage experience ($\vartheta_{i,r}$)), (3) decay parameters related to purchase utility (δ_i) and return cost (R_i), and (4) coefficients related to the RFM measures. We allow all parameters to be individual specific to account for unobserved heterogeneity.

We can identify shopping trip fixed cost by the frequency of trips consumers made during the sample period. Those who incur higher fixed costs are likely to make fewer trips, when holding all else constant. As Fig. 2 shows, the number of shopping trips varies considerably, and these variations enable us to identify the heterogeneity of fixed cost.

Next, recall that purchase utility depends on both the realization of perceived value and price. The average prices consumers paid for all purchases identify parameter d_i. Those who hold the firm's products in high regard, or have higher perceived value d_i, are more likely to purchase more expensive products. The variance of perceived value signals in-store (ϑ_i) is identified with the variance in prices paid for different products across all shopping trips. If consumers purchase products with highly different prices, it is likely that they are receiving entirely different perceived value signals during these trips. Consequently, the variance of perceived value signals obtained in-store will be greater. We identify the decay parameter related to purchase utility (δ_i) from the average number of products purchased in each trip. The closer δ_i is to 1, the more linear consumers' purchase utility is and the less decline in utility occurs from additional purchases. Therefore, consumers whose δ_i is closer to 1 are more likely to purchase more in each shopping trip. Heterogeneity in decay parameter could be then inferred from the variations in the number of purchases per trip.

In our model, we do not estimate the price coefficient. In general, identifying the price coefficient requires relating repeated purchases of the same stock-keeping unit to the price variations. Such patterns of repeated purchases, however, are not prevalent in our empirical context of personal care products. Therefore, we set the price coefficient (α_i) to 1; researchers facing similar challenges have also used this specification (Chen and Yao 2016; Nevo et al. 2016). However, such a normalization will scale all remaining parameters at the same time and thus will not influence the results of our policy simulations.[7]

Last, similar to parameter ϑ_i, we identify parameter $\vartheta_{i,r}$, or the variance of perceived value signals from usage experience, by the variance of prices of products being returned. Return cost (R_i) is the portion of products being returned by each consumer. In our data, roughly 11% of the total purchases are returned, which suggests the presence of return costs. To infer the coefficients related to the RFM measures, we relate interperiod variations to the variation of the RFM measures. Table 3 summarizes our identification strategy.

[7] We also checked the robustness of our counterfactual results under the model estimates with different levels of the normalized price coefficient. Results are not sensitive to these alternative specifications.

Results

In this section, we present inferences from the parameter estimates based on our proposed structural model. Using our estimation results, we then conduct a few policy simulations and discuss the managerial implications of the results.

Model estimates

Table 4 presents the parameter estimates of our proposed structural model (posterior means and credible intervals for statistical inferences). First, we note that consumers incur a considerable cost to initiate a shopping trip, with the fixed cost coefficient (ν_i) estimated as −32.26. Our model estimates also suggest that consumers incur a considerable marginal cost, estimated as −17.21, to return each purchased item. We explore the implications derived from these two types of costs and provide several alternative strategies for the marginal cost of product returns.

Second, regarding the parameters related to purchase and return utility, in our model we allow consumers to obtain information about their perceived value of the firm's products from both in-store observations and consumption experiences. In particular, we assume that both types of information are randomized draws from a log-normal distribution with mean (d_i), capturing consumers' true perceived value. Therefore, both types of information are unbiased representatives of consumers' true perceived value. However, we allow the standard deviation of the distribution of the two types to differ. Our estimation results suggest a larger standard deviation related to the information about product values obtained in-store, which is reasonable in the retail context. A larger standard deviation implies that perceived value signals obtained in-store can vary considerably across different shopping occasions. As discussed earlier, perceived value signals obtained in-store could be driven by how products are displayed and presented in stores. As it is common for retailers of the focal category to frequently, or at least seasonally, update product displays in-store, impressions of products in different displays could lead to different perceived product values.

Last, we turn to the utility of making no shopping trips, which is determined by a consumer's RFM measures. Recall that the recency measure captures the number of weeks between the current purchase and a consumer's most recent purchase; thus, a higher recency measure suggests lower engagement from the focal consumer as he or she has not shopped at the retailer for a while. Given this, his or her utility of making no shopping trips should be higher. We can rationalize such a conjecture if the coefficient associated with the recency

Table 3 Identification

	Notation	Source of identification
Purchase utility	d_{ie}	Average prices consumers paid for all purchases
	δ_i	Average number of products purchased during each trip
	v_i	Frequency of trips consumers made during the sample period
Return utility	$d_{ig,\,r}$	Variance of prices of products being returned
	R_i	Portion of products being returned by each consumer

measure is positive ($\alpha_i^r = 0.95$). Meanwhile, higher frequency and monetary values suggest that the consumer is highly engaged with the focal firm. Consequently, he or she derives lower utility from making no shopping trips ($\alpha_i^f = -0.15; \alpha_i^m = -0.41$).

Effects of product returns[8]

What are the effects of product returns? As discussed previously, causally quantifying these effects using archival data is challenging because of potential selection biases. Instead, firms need to add exogenous variations to their return policies (i.e., through field experiments); however, such exogenous variations are costly to achieve in reality. One advantage of structural models is the possibility to simulate consumer behaviors under hypothetical scenarios, by altering certain modeling components, rather than actually changing firms' return policies. Similar to field experiments, all implemented variations are exogenous (as they are implemented by researchers), and thus the results of the policy simulations enable us to causally quantify the impact of product returns. In this section, we elaborate on the design of our counterfactuals in quantifying the option value and shopping trip effect of product returns.

Option value effect We first discuss how we implement the counterfactuals in quantifying the effects of product returns through the option value (Anderson et al. 2009). Specifically, this option value captures the impact of future return utility on current-period purchase decisions. In our model, $U_{it}^b + U_{it}^r + v_i + \beta.EV\left(x_{i,t+1} \,|E_{it}\,, G_{it}\right)$ determines a consumer's decision in the current period, where $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$ is the expected utility associated with future shopping trips. As discussed previously, parametrically $EV\left(x_{i,t+1} \,|E_{it}\,, G_{it}\right) = \ln\left\{\sum_{\{E,G\}}\exp\left(U_{i,t+1}^b + U_{i,t+1}^r + \beta.EV\left(x_{i,t+2}\,|E_{i,t+1}\,, G_{i,t+1}\right)\right) + \exp\left(f\left(r_{i,t+1}\,, f_{i,t+1}\,, m_{i,t+1}\right) + \beta.EV\left(x_{i,t+2}\,|E_{i,t+1}\,, G_{i,t+1}\right)\right)\right\}$. We therefore can transfer such an optional value into the term $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$. A higher expected return utility $U_{i,t+1}^r$ increases $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$, which in turn influences the current-period purchase decision.

To simulate the scenario without the option value of product returns, we set the utility of return $\left(U_{i,t+1}^r\right)$ to 0 when computing $EV(x_{it}|\,E_{it}, G_{it})$. With this specification, future return utility no longer influences the magnitude of $EV(x_{it}|\,E_{it}, G_{it})$ or the current-period purchase decision. This specification therefore nullifies the option value effect of product returns. Accordingly, we can revise the parametric specification of $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$ as $EV\left(x_{i,t+1}\,|E_{it}\,, G_{it}\right) = \ln\left\{\sum_{\{E,G\}}\exp\left(U_{i,t+1}^b + \beta.EV\left(x_{i,t+2}\,|E_{i,t+1}\,, G_{i,t+1}\right)\right) + \exp\left(f\left(r_{i,t+1}\,, f_{i,t+1}\,, m_{i,t+1}\right) + \beta.EV\left(x_{i,t+2}\,|E_{i,t+1}\,, G_{i,t+1}\right)\right)\right\}$.

Note that in this counterfactual, we want to quantify the option value effect only; therefore, we can only change consumers' expectation of future return opportunities through $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$. The other elements that govern consumers' current-period decisions (the flow utility, or $U_{it}^b + U_{it}^r + v_i$) remain unchanged in this counterfactual. In summary,

Table 4 Structural estimation results

Parameter	Posterior mean	Credible interval	
		Lower bound	Upper bound
Fixed cost (v_i)	−32.26	−37.91	−26.79
Return cost (R_i)	−17.21	−20.00	−14.52
Perceived value			
Mean (d_i)	7.81	6.38	9.29
Standard deviation (ϑ_i)	1.20	1.03	1.26
Standard deviation ($\vartheta_{i,\,r}$)	0.87	0.78	0.99
Decay (δ_i)	0.48	0.15	0.82
No visit (RFM)			
Recency (α_i^r)	0.95	0.01	1.89
Frequency (α_i^f)	−0.15	−0.27	−0.05
Monetary value (α_i^m)	−0.41	−0.79	−0.06
Model fit			
Log-marginal density	−2139.6		
Hit rate	0.91		

[8] In all simulations described in this and the next sections, we update the state variables (number of products eligible for returns and RFM measures) on the basis of predicted consumer purchase and return decisions under each scenario.

this counterfactual simulates the scenario in which the option value is nullified, as the further expected return utility no longer influences current-period purchase decisions through $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$. By contrast, the shopping trip effect is still maintained under this counterfactual through U^r_{it}.

This hypothetical scenario is challenging to achieve in reality, as is quantifying the shopping trip effect of product returns, because it involves manipulating consumers' expectations [$EV(x_{i,\,t+1}|\,E_{it}, G_{it})$]. Such changes, however, can be easily achieved through our proposed structural model, thus highlighting the importance of our proposed structural model in understanding the impact of product returns.[9]

We compare a series of metrics (number of shopping trips, gross purchase, net purchase after return, and revenue) in each counterfactual scenario with the baseline scenario (in which $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$ still depends on the expected future return utility $U^r_{i,t+1}$). As the net purchase considers a consumer's product returns, a change in this metric therefore accounts for both the positive effects and negative effects (e.g., loss in sales) of product returns. These metrics are aligned with those used to assess the performance of retailers. Table 5 reports the results.

As the first row of Table 5 shows, in the absence of a return's option value effect, consumers are disincentivized to make more purchases. We find that both gross purchase and net purchase decline by 24.87% and 20.68%, respectively, under the first counterfactual scenario. The number of shopping trips also declines by 2.06%. These observations, particularly the decline in net purchase, demonstrate the option value effect of product returns.

Shopping trip effect In our proposed (baseline) model, the utility derived from each shopping trip is $U^b_{it} + U^r_{it} + v_i + \beta.EV\left(x_{i,t+1}\,|E_{it}\,, G_{it}\right)$, where U^r_{it} is the return utility in the current period. As discussed previously, the shopping trip effect depends on the magnitude of U^r_{it}. That is, a higher return utility will incentivize consumers to make additional shopping trips in the current period. Accordingly, to nullify the shopping trip effect, we could set U^r_{it} to 0, such that the current-period return utility no longer influences consumers' shopping trip utility and, thus, the decision to make a shopping trip. Accordingly, we can revise the utility derived from each shopping trip as $U^b_{it} + v_i + \beta.EV\left(x_{i,t+1}\,|E_{it}\,, G_{it}\right)$.

When simulating consumer behaviors under this counterfactual scenario, we only nullify the shopping trip effect by setting U^r_{it} to 0; the option value effect is preserved. In particular, the term $EV(x_{i,\,t+1}|\,E_{it}, G_{it})$ under this counterfactual depends on both future purchase and return utility, or EV

$$\left(x_{i,t+1}\quad\quad|E_{it}\quad\quad,G_{it}\right)=\ln\left\{\sum_{\{E,G\}}\exp\left(U^b_{i,t+1}+U^r_{i,t+1}+\beta.EV\left(x_{i,t+2}\;|E_{i,t+1}\;,G_{i,t+1}\right)\right)+\exp\left(f\left(r_{i,t+1}\quad,f_{i,t+1}\quad,m_{i,t+1}\right)+\beta.EV\left(x_{i,t+2}\quad|E_{i,t+1}\right.\right.\right.$$
$$\left.\left.\left.,G_{i,t+1}\right)\right)\right\}.$$

We report the magnitude of this shopping trip effect in the second row of Table 5. When returns are not allowed (under the second counterfactual), the number of total shopping trips declines by 4.99%, with a decrease in the gross purchase and net purchase of 11.12% and 0.78%, respectively. Our simulation results therefore validate our conjecture that product returns can effectively incentivize consumers to make additional shopping trips. While these trips are made with the original intent to return, consumers might end up making additional purchases. Doing so leads to an increase in both gross and net purchases.

To summarize, the results from the two policy simulations show the value of product returns to firms, through the option value and shopping trip effects. More importantly, we further show that both effects can promote consumer engagement on the dimensions of net purchases and shopping trips. These observations suggest that managing product returns purely as a cost is suboptimal for retailers; instead, retailers should consider alternative marketing strategies to leverage the potential benefits associated with product returns.

Targeted reduction of return cost

Motivated by the findings about product returns, we now explore possible approaches for retailers to leverage marketing opportunities associated with product returns. According to Abbey et al. (2018), targeting strategies with product returns are still fairly limited among retailers, with firms often applying a one-size-fits-all approach to their entire customer base. Toward this end, we propose and evaluate a policy of targeted reduction toward return cost. As noted previously, consumers incur sizable return costs. As our previous analysis demonstrates, the high return cost can be suboptimal for retailers because it reduces the expected return utility and, thus, consumers' option value from return. In turn, this lower option value from returns can disincentivize purchases.

This conjecture is further strengthened when consumers have high variance associated with their match value signals. Again, the variance of match value signals captures the consistency of signals obtained on different occasions. With a high variance of these signals, consumers could obtain entirely different signals from in-store and postpurchase usage experiences (e.g., signals from usage may be considerably lower than those observed in-store). Such discrepancy could disincentivize purchases to avoid unexpected low-match

[9] The goal of simulations in this section is to quantify the impact of product returns rather than to propose alternative managerial strategies. Thus, whether the proposed scenario could be achieved in reality is less of concern. In the next two sections, we shift our focus to explore returns-based targeting strategies, in which we discuss in detail how retailers can implement each proposed strategy in reality.

Table 5 Effect of returns and return-based targeted price promotions

	Shopping trips	Gross purchase[a]	Net purchase[b]	Revenue[c]
Effect of Return				
Option value	−2.06	−24.87	−20.68	−21.68
Shopping trips	−4.99	−11.12	−0.78	−1.91
Targeted Reduction				
Return cost	1.65	7.67	7.52	7.13
Price	1.30	11.41	11.51	11.64

Note: All numbers are shown as % change compared with the benchmark policy

[a] Number of products purchased (including those being returned later)

[b] Number of products purchased (subtracting those being returned later)

[c] Revenue is computed as net spending from consumers

values (e.g., possible allergies) from usage, even if consumers observe a high-match signal in-store. The possibility to return could counteract the adverse impact of the high variance (i.e., consumers could still return the products if match values turn out to be much lower from usage experience). A reduction in return costs for these consumers could therefore be beneficial for retailers by further increasing both the option value and consumer spend.

To assess the potential benefit of such a strategy, we consider a policy in which the retailer offers a targeted 10% reduction of the return cost (by changing parameter R_i) to consumers whose variance of perceived value signals ($\vartheta_i, \vartheta_{i,\ r}$) is higher than the populational mean (among all consumers). Retailers such as Target have adopted such strategies (The Krazy Coupon Lady 2019b). To our knowledge, our focal retailer does not have similar policies. In practice, identifying the focal type of consumers can be achieved by either estimating our proposed model or using some empirical measures. Recall that the variance of perceived value signals captures both store and product heterogeneity. Therefore, for a given consumer, his or her individual variance of perceived value signals could be inferred from the store where he or she usually shops and the product assortment usually purchased from the focal retailer.

When assessing the effectiveness of our proposed targeting strategy, we compare our strategy with a benchmark policy in which a reduction is offered to a randomly selected set of consumers (we keep the number of these consumers equal to the number of consumers selected in our proposed targeting strategy). In contrast with our proposed strategy, the benchmark strategy is untargeted, and thus such a comparison enables us to highlight the gain from our proposed targeting strategy. The third row of Table 5 reports the results under this counterfactual. Not surprisingly, we find that this targeted strategy promotes shopping trips by 1.65%, gross purchase by 7.67%, and net purchase by 7.52%.

Targeted price promotion on return trips

As our previous analysis indicates, returns can incentivize consumers to make additional shopping trips. Attracting store traffic has become increasingly challenging for retailers, given the intensified competition. Retailers therefore should promote purchases during these additional shopping trips even though these trips are made with the original intent of product returns. In this section, we consider a counterfactual scenario in which the retailer offers a 10% price discount to consumers who return products in-store. Such a promotion strategy has received increasing attention from industry practice. Kohl's, for an instance, has begun offering price coupons to all consumers who make in-store returns of products purchased on Amazon.com (The Krazy Coupon Lady 2019a). However, understandings regarding the effectiveness of this strategy are still at an early stage in academic research. With our proposed structural model, we could evaluate this strategy through policy simulations.

Similar to the previous policy, price discounts could promote conversion even when offered randomly. To obtain fair policy inferences, we compare the performance of our proposed targeted price discount with a benchmark policy in which a 10% price discount is offered to a subset of randomly selected shopping trips.[10] As the last row of Table 5 shows, our proposed targeted price promotion outperforms the benchmark policy (randomized price promotion) in terms of shopping trip frequency (by 1.30%), gross purchase (by 11.41%), and net purchase (by 11.51%).[11]

[10] We equate the number of trips being targeted with this price discount between the benchmark and the proposed policy.

[11] When running the simulations, we also adjust consumers' price expectations due to the additional price promotions. By doing so, the results reported from this policy simulation account for the potential strategic product return behaviors (e.g., consumers might purposefully make product returns to receive the targeted price discount).

Conclusions

This study examines the effect of product returns on both purchases and shopping trips using a dynamic structural model. Our structural model characterizes two effects of product returns: (1) the option value effect and (2) the shopping trip effect. In that regard, our proposed model extends extant literature by examining a consumer's multi-dimensional (i.e., in terms of both purchases and shopping trips) responses to product returns. Using a unique dataset from a retailer of personal care products, we demonstrate that both effects of product returns can promote consumer engagement on the dimensions of net purchases and shopping trips. In other words, allowing product returns is ultimately in retailers' best interests.

Leveraging one key advantage of the structural model, we further explore possible approaches for retailers to leverage marketing opportunities associated with product returns, using policy simulations. Particularly, we experiment two potential targeting strategies: (1) targeted reduction of return cost and (2) targeted price promotion on return trips. Our simulation results suggest that both policies could effectively promote net purchases.

Insights derived from this paper could be practically useful to retailers for the following two reasons. First, the two strategies proposed in this paper directly address the key limitation in current industry practices; particularly, the absence of targeting strategies associated with product returns (Abbey et al. 2018). Second, it is also important to note that policy simulations, constructed based on our proposed structural model, allow retailers to assess these managerial strategies without incurring additional financial costs. For instance, to evaluate the targeted price promotion strategy, our policy simulation does not require retailers to offer any actual coupons to consumers in reality. Given that retailers are not aware of the exact impact of this targeting strategy *ex-ante*, both magnitude and direction-wise, assessing such a strategy in reality (e.g., through a field experiment) could imply considerable opportunity costs. Our proposed structural model as well as the policy simulations alleviate such uncertainty and financial costs associated with the assessment of various managerial strategies.

Future research can extend our study in at least two ways. First, research could extend our proposed model to examine product returns with online and omnichannel retailers (e.g., Narang and Shankar 2019). Most omnichannel retailers allow consumers to return online purchases to stores (Appriss Retail 2018); consequently, the shopping trip effect of product returns is still applicable to these retailers. Even if visits to physical stores are not necessary for product returns (e.g., returning by mail), consumers might still need to visit the firm's website for purposes such as printing the return label and/or order receipts. Similar to brick-and-mortar stores, these additional website visits might lead to additional purchases. In that regards, the shopping trip effect and targeting strategies associated with product returns, identified with brick-and-mortar store in this paper, might be applicable to online and omni-channel retailers as well. With the growing availability of consumer-level data in online and omnichannel retail, we hope our proposed model could be extended to better understand the impact of product returns with different types of retailers.

A second possible extension of our model is to include return experience (e.g., whether consumers are satisfied with the return process or return policy). As Ertekin (2018) documents, a positive return experience could promote purchases in both the short and long run. Our framework can accommodate such an extension given the increasing availability of data on consumers' return experiences (e.g., through consumer surveys).

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Appendix: Implementation

Computational details

Here, we specify our implementation of the IJC framework. We denote the parameter set as θ_i, which includes (1) shopping trip fixed cost (v_i), (2) parameters related to purchase utility (i.e., mean (d_i) and variance of match value signals in-store (ϑ_i) and from usage experience ($\vartheta_{i,r}$)), (3) decay parameters related to purchase utility (δ_i) and return cost (R_i), and (4) coefficients related to the RFM measures. Individual-specific parameters come from $\{\theta_i\} \sim N\left(\widetilde{\theta}, \sigma_\theta^2\right)$.

Imai et al. (2009) propose a less computationally intensive approach to approximate value function using realizations of the past 100 iterations ($N = 100$). Suppose that we are at iteration r and have accumulated $H^r = \left\{ \left\{ x_i^k,\ \widetilde{V}^k\left(x_i^k;\theta_i^k\right) \right\}_{i=1}^{I} \right\}_{k=r-N}^{r-1}$ through past iterations, where k denotes a given iteration before the current r^{th} iteration, x_i^k and θ_i^k are the set of state variables and parameters stored at each past iteration (to be discussed subsequently), and $\widetilde{V}^k\left(x_i^k;\theta_i^k\right)$ is the value function associated with the stored state variables and parameters at each past iteration. Given that our model accommodates individual-specific parameters, in theory it requires us to store all state variables, parameters, and realized value functions for all consumers across all past iterations, which is computational costly. Thus, we adopt Imai et al.'s (2009) procedure to store one realization from one randomly selected consumer during each iteration. This

simplification reduces the requirement for computer memory considerably. Within the rth iteration, the estimation procedure is extended as follows:

Step 1: Update * $\{\theta_i\}$

1. We first draw a new set of parameter values of $\{\theta_i\}$ for each consumer using random walk. We denote the candidate draw as $\{\theta_i^c\}$.
2. To assess the value function, we must compute the $EV(x_{i,t+1})$. Following Imai et al. (2009), we can compute $EV(x_{i,t+1})$ using all past state variables, parameters, and realizations of value functions, as follows:

$$\widetilde{EV}\left(x_{i,t+1}\right) = \sum_{k=r-N}^{r-1} \widetilde{V}^k * \frac{K_{h\theta}\left(\theta_i^c-\theta_i^k\right)K_{hINV}\left(INV_i-INV^k\right)K_{hRFM}\left(RFM_i-RFM^k\right)f\left(p_i-p^k\right)f\left(d_{ig,r}-d_{ig,r}{}^k\right)}{\sum_{k=r-N}^{r-1} K_{h\theta}\left(\theta_i^c-\theta_i^k\right)K_{hINV}\left(INV_i-INV^k\right)K_{hRFM}\left(RFM_i-RFM^k\right)f\left(d_{ig,r}-d_{ig,r}{}^k\right)},$$

where INV_i is the total number of products available for return and RFM_i is the set of RFM measures. Given consumers' current-period purchase and return decisions, we can compute INV_i and RFM_i deterministically; K_{hINV} and K_{hRFM} are their kernel density functions accordingly. The other two state variables—average price of purchased products, p_i, and match value from usage experience $d_{ig,r}$—evolve stochastically.[12]

3. To compute the likelihood function, we must also integrate out the random match values. However, the complexity of our proposed model prevents us from analytically integrating out these two random match values. Following prior research (e.g., Erdem and Keane 1996), we draw 100 sets of random match values from their distributions (as specified in the main body of the paper) to integrate out the likelihood function.
4. With the likelihood obtained with θ_i^c and θ_i^{r-1}, we determine whether or not to accept the candidate parameter draw for a given individual.

Next, we repeat Step 1 to update individual-level parameters for every consumer.

Step 2: Update $\widetilde{V}^r\left(x_i^r;\theta^r\right)$

5. The last step for iteration r involves computing the pseudo-value function $\widetilde{V}^r\left(x_i^r;\theta_i^r\right)$, which can be done as follows:
   - i Make one draw from the state space;
   - ii Compute the expected future value;
   - iii Apply the Bellman operator once.
6. Proceed to the next iteration

[12] The general idea of the IJC approach is to approximate the value functions by the average of their past realizations, as weighted by the difference between the state variables at the current iteration for the focal consumer and the state variables at past iterations for randomly selected consumers.

State variables

Here, we now discuss the specification and law of motion associated with our state variables.

Number of products available for return We specify the number of products available for return as $INV_{it} = INV_{i,t-1} + E_{it} - G_{it}$. For simplicity of notation, we denote $INV_{i,t-1}$ as the number of products purchased before the current trips that are still within the window of return, E_{it} as the number of newly purchased items, and G_{it} as the number of items returned during the current visit. As we do not observe consumers' purchases from the very beginning, our proposed model is subject to a left-truncation issue (i.e., we do not observe INV_{i0}). To address this issue, we first estimate a preliminary model, which has the same structure of our proposed model but with the assumption that INV_{i0} equals 0 (i.e., consumers have no items that are available to return at the beginning). With the model estimates, we then forward-simulate consumers' purchases and returns 100 times randomly draw a set of initial INV_{i0} to estimate our proposed model.

RFM measures Again, in our model, we include the following RFM measures: (1) the number of weeks from the most recent purchase (recency), (2) the number of shopping trips with purchases made in the recent quarter (frequency), and (3) the number of net spending in the recent quarter (monetary). Similar to inventory, all RFM measures are subject to left-truncation. We therefore use the same approach, as described previously, to handle the left-truncation issue with RFM measure.

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